

# MINDMASTER THUNDERSTUDY ACADEMY

## OFFICIAL MARKING SCHEME / ANSWER KEY (2026-27)

SUBJECT: PHYSICAL EDUCATION (THEORY) | MAX MARS: 70

### SECTION – A (MULTIPLE CHOICE QUESTIONS - 1 MARK EACH)

1. (C) Suryabhedan
2. (C) 1989
3. (C) Roughage
4. (A) Law of Inertia
5. (C) Red, blue, yellow and green
6. (A) Bhujangasana
7. (B) Arm Curl Test
8. (B) Opposite direction

### SECTION – B (VERY SHORT ANSWER TYPE - 2 MARKS EACH)

#### 19. Second Class Lever:

- **Definition:** In a second-class lever, the load (resistance) is located between the fulcrum and the force (effort).
- **Mechanical Advantage:** There is always a mechanical advantage as the force arm is always greater than the resistance arm.
- **Example in Sports:** Raising the body on the toes (Plantar flexion) or a full-body push-up.

#### 20. Macro vs. Micro Nutrients:

**Macro Nutrients:** Required in large quantities; have calorific value; consist of energy and body-building elements (e.g., Carbohydrates, Fats, Proteins, Water).

**Micro Nutrients:** Required in small quantities; have no calorific value; consist of protecting elements (e.g., Vitamins and Minerals).

#### 21. Corrective Measures:

- **Bow Legs:** Walk on the inner edge of the feet; perform Garudasana or Ardh Chakrasana; use braces or modified shoes.
- **Round Shoulders:** Perform Chakrasana and Dhanurasana; hold the horizontal bar; perform chest stretches or Bhujangasana.

## 22. BMI Calculation:

**Formula:**  $\text{Weight (kg)} / [\text{Height (m)}]^2$

**Calculation:**  $72 / (1.68 * 1.68) = 72 / 2.8224$

**Result:** 25.51 (Overweight category)

## 23. Benefits for CWSN (Children with Special Needs):

- **Physical:** Improves motor skills, flexibility, and cardiovascular health.
- **Psychological:** Builds confidence, reduces anxiety, and enhances focus.
- **Social:** Enhances teamwork, communication skills, and promotes inclusion.
- **Emotional:** Elevates mood and develops resilience.

## 24. Friction in Sports:

**Definition:** A force acting over the area of contact between two surfaces in the direction opposite to motion.

**Types:** (1) Static Friction (Take off blocks). (2) Sliding Friction (Ice skating). (3) Rolling Friction (Cricket ball). (4) Fluid Friction (Swimming/Cycling).

## SECTION – C (SHORT ANSWER TYPE - 3 MARKS EACH)

## 25. Talent Identification vs. Talent Development:

- **Identification:** Recognizing potential at an early age using physiological and psychological testing and standard norms.
- **Development:** Providing a suitable environment using systematic and scientific long-term processes to realize potential.

## 26. Vitamin D and Vitamin K:

- **Vitamin D:** Absorption of calcium/phosphorus; bone strength. Source: Sun, milk.
- **Vitamin K:** Blood clotting; energy production in mitochondria. Source: Leafy greens, cabbage.

## 27. Committees:

- **Technical Committee:** Handles official arrangements, match conduct, and fixtures.
- **Marketing Committee:** Manages advertisements, broadcasting, and securing sponsorships.

## 28. Physiological Changes due to Ageing:

- **Muscular:** Decline in strength and mass.
- **Skeletal:** Loss of bone density (osteoporosis).
- **Cardiovascular:** Decrease in cardiac output and stroke volume.
- **Nervous:** Diminished memory and slower reaction.

## 29. Hostile vs. Instrumental Aggression:

- **Hostile:** Primary goal is to cause harm/injury; accompanied by anger.
- **Instrumental:** Aggressive behavior to achieve a goal (like winning) without primary intent to harm.

## 30. Flexibility improvement methods:

Flexibility is the ROM around a joint. Methods includes: (1) Ballistic (swinging/momentum). (2) Static (hold stretch). (3) PNF (Contract-Relax).

## SECTION – D (CASE STUDIES - 4 MARKS EACH)

### 31. Fixture Analysis (11 Teams):

- (i) matches =  $N - 1 = 11 - 1 = 10$  matches
- (ii) Byes = Power of 2 (16) - 11 = 5 Byes
- (iii) Rounds = 4 rounds (as  $2^4 = 16$ )
- (iv) Seeding = Placing top teams in different halves to avoid early meetings.

### 32. Biomechanics:

(i) Law: Law of Acceleration / Inertia. (ii) Friction: Opposes motion (Rolling). (iii) Relation: Force = Mass \* Acceleration. Heavier objects need more force.

### 33. CWSN Logos:

(i) Logo identified as Special Olympics. (ii) Founder: Eunice Kennedy Shriver. (iii) Motto: "Spirit in Motion".

## SECTION – E (LONG ANSWER TYPE - 5 MARKS EACH)

### 34. Yoga for Hypertension/Asthma:

**List:** Tadasana, Bhujangasana, Gomukhasana, Shavasana.

**Procedure (Bhujangasana):** Lie prone, hands near shoulders, inhale and lift chest like a cobra.

**Contraindications:** Avoid in cases of abdominal surgery, hernia, or pregnancy.

### 35. Circuit Training:

**Definition:** Stations-based training with specific dosage and minimal rest. **Planning:** Coach sets 6-10 stations (Push-ups -> Squats -> Crunches -> Skipping -> Shuttle run) for full-body conditioning.

### 36. Rikli and Jones Fitness Test:

Standardized test for Seniors. (1) Chair Stand (Lower body strength). (2) Arm Curl (Upper strength). (3) Chair Sit & Reach (Flexibility). (4) Back Scratch (Shoulder flexibility). (5) 8-foot Up & Go (Agility). (6) 6-Min Walk (Endurance).

### 37. Fixture Drawing (21 Teams):

$N = 21$ .  
Upper Half = 11, Lower Half = 10.  
Byes =  $2^5 - 21 = 32 - 21 = 11$  Byes.  
Matches = 20. Rounds = 5.

Draw steps: Place byes bottom-up, top-down sequence. Match remaining teams in rounds.